

## Project Workflow: Week 1

**Project:** EasyStego: A Drag-and-Drop Image Steganography Tool

**Phase:** Project Planning & Concept Finalization

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**Course:** Software Development I (CSE 2216)

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### 1. Objective

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The primary goal of Week 1 was to establish a robust foundation for the EasyStego project. This included:

- Defining the project scope and functional boundaries for the steganography tool.
- Selecting an optimal technology stack for desktop deployment and cross-platform support.
- Conceptualizing the UI/UX design to align with a simple, non-technical user experience.

### 2. Detailed Task Breakdown

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**Feature Identification:** Researched existing steganography tools and techniques to finalize the core feature set — LSB-based text embedding, drag-and-drop image loading, real-time capacity calculation, and message extraction.

**Technical Stack Selection:** Selected Java Swing for its mature component library, native drag-and-drop support via TransferHandler, and platform independence across Windows, macOS, and Linux.

**UI/UX Design:** Designed a two-tab interface (Hide Message / Reveal Message) with a dashed drop-zone for image input, a Windows-blue accent color (#0078D7), and a clean Segoe UI typeface for readability.

**Architectural Setup:** Designed a modular single-class architecture (extending JFrame) with a custom inner TransferHandler class to cleanly separate drag-and-drop logic, UI construction, and LSB encoding/decoding logic.

### 3. Deliverables

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- Project Scope Document: a formal outline of EasyStego's features and limitations.
- UI Wireframe Sketches: conceptual layout of the drop zone, message panels, and tabbed interface.
- Color Palette Definition: a finalized accent color and drop-zone styling for the Swing UI.
- Project Timeline: a structured 10-week development roadmap.

## 4. Tools & Resources

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### **Development Environment:**

- Java JDK 11+
- Visual Studio Code

### **Reference Materials:**

- Java Swing and AWT documentation ([javax.swing](#), [java.awt.datatransfer](#))
- [javax.imageio.ImageIO](#) for PNG read/write operations